

INDIAN INSTITUTE OF TECHNOLOGY PALAKKAD

DS3010 INTRODUCTION TO MACHINE LEARNING

BREAST CANCER DIAGNOSIS USING MACHINE LEARNING

PROJECT PROBLEM STATEMENT

Submitted by:

Kashishdeep Kaur (142401020)

Rishitha (142401022)

Greeshma (142401024)

Joshitha (142401040)

Instructor: Dr. Swapnil Hingmire



INDIAN INSTITUTE
OF TECHNOLOGY
PALAKKAD

1 Problem Statement

Breast cancer is one of the major health challenges worldwide, and early and accurate diagnosis is important for appropriate clinical decision-making. The objective of this project is to develop and evaluate machine learning approaches for distinguishing between **malignant** and **benign** breast masses using numerical features extracted from digitized fine-needle aspirate (FNA) images.

The project will use the **Breast Cancer Wisconsin (Diagnostic)** dataset available from the UCI Machine Learning Repository. The dataset contains measurements describing characteristics of cell nuclei present in digitized images of breast mass samples. These measurements can be used as input features for building classification models.

The central problem addressed in this project is:

Can machine learning models accurately classify breast masses as malignant or benign based on quantitative characteristics extracted from cell-nucleus images?

The project will involve exploring the dataset, performing appropriate data preprocessing and exploratory data analysis, selecting relevant features, training classification models, and evaluating their performance using suitable classification metrics.

2 Goal of the Project

The primary goal of this project is to develop a reliable machine learning-based classification system for predicting whether a breast mass is **malignant** or **benign** using the numerical features provided in the Breast Cancer Wisconsin (Diagnostic) dataset.

The specific objectives are:

1. To understand and explore the characteristics of the breast cancer diagnostic dataset.
2. To perform exploratory data analysis and identify patterns, distributions, correlations, and relationships among the features.
3. To preprocess the data appropriately for machine learning models.
4. To investigate the importance and relevance of the available features for breast cancer classification.
5. To develop and compare suitable supervised machine learning classification algorithms.
6. To evaluate the models using appropriate metrics such as accuracy, precision, recall, F1-score, and confusion matrix.
7. To identify a model that provides effective classification performance while considering the importance of correctly identifying malignant cases.

3 Potential Domains to Explore

The project primarily falls under the following domains:

- **Machine Learning:** Supervised classification techniques will be used to distinguish between malignant and benign cases.
- **Healthcare and Medical Diagnosis:** The project explores the application of computational methods to breast cancer diagnosis.
- **Data Science:** Data preprocessing, exploratory data analysis, feature analysis, visualization, and model evaluation will form important parts of the project.
- **Medical Data Analysis:** Relationships between cell-nucleus characteristics and diagnostic outcomes will be investigated.
- **Explainable Machine Learning:** Feature importance and model interpretability may be explored to understand which characteristics contribute most to the predictions.

4 Dataset

The project will use the following dataset:

- **Dataset Name:** Breast Cancer Wisconsin (Diagnostic)
- **Source:** UCI Machine Learning Repository
- **Dataset ID:** 17
- **Number of Instances:** 569
- **Number of Features:** 30 real-valued features
- **Target Variable:** Diagnosis
- **Classes:**
 - M - Malignant
 - B - Benign
- **Missing Values:** None

The 30 numerical features are derived from ten characteristics of cell nuclei: radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension. For each characteristic, measurements corresponding to the mean, standard error, and worst/largest value are provided.

The dataset also contains an identification number, which will be treated as an identifier rather than a predictive feature.

5 Potential Machine Learning Approaches

The following approaches may be explored and compared as part of the project:

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)
- Naive Bayes
- Other suitable classification techniques, if required

The final selection of models will depend on the results obtained during exploratory analysis and experimentation.

6 Model Evaluation

Since the problem is a binary classification task, model performance will be evaluated using multiple metrics, including:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Particular attention will be given to **recall for the malignant class**, since incorrectly classifying a malignant case as benign can have serious consequences in a medical diagnostic setting.

7 Expected Outcome

The expected outcome of the project is a comparative analysis of different machine learning classification techniques for breast cancer diagnosis. The project aims to determine which approaches can effectively distinguish malignant and benign cases using the available diagnostic features.

The analysis is expected to provide insights into:

- The characteristics of the dataset and its class distribution.
- The most informative features for classification.

- The effectiveness of different machine learning algorithms.
- The trade-offs between different evaluation metrics.
- The suitability of machine learning as a computational aid for breast cancer classification.

The resulting models will be considered as analytical and educational tools rather than replacements for professional medical diagnosis.

8 Dataset Reference

Wolberg, W., Mangasarian, O., Street, N., & Street, W. (1993). *Breast Cancer Wisconsin (Diagnostic) [Dataset]*. UCI Machine Learning Repository.

DOI: <https://doi.org/10.24432/C5DW2B>

Dataset URL: <https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic>